

BENCHMARK: NON-DETECTION

Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2456348.85139 +/- 0.00054 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.026 +/- 0.039                   |
| Transit depth (Rp/Rs)^2                           | 0.067 +/- 0.2 [%]                 |
| Orbital Inclination (inc)                         | 86.56 +/- 3.1                     |
| Airmass coefficient 1 (a1)                        | 1.053 +/- 0.017                   |
| Airmass coefficient 2 (a2)                        | -0.038 +/- 0.059                  |
| Scatter in the residuals of the lightcurve fit is | 1.63 %                            |
| Best Comparison Star                              | #3 - [22, 206]                    |
| Optimal Aperture                                  | 2.72                              |
| Optimal Annulus                                   | 6.06                              |
| Transit Duration (day)                            | 0.0511 +/- 0.0063                 |

Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.026 ± 0.039 vs 0.1594 (deviation 3.19σ) |
| Duration fit vs published | 0.0511 d vs 0.0512 d (deviation 0.02σ)    |
| Mid-time O–C              | 1.1 min                                   |
| Depth SNR                 | 0.6                                       |
| Residual scatter          | 1.63 %                                    |

Light curve

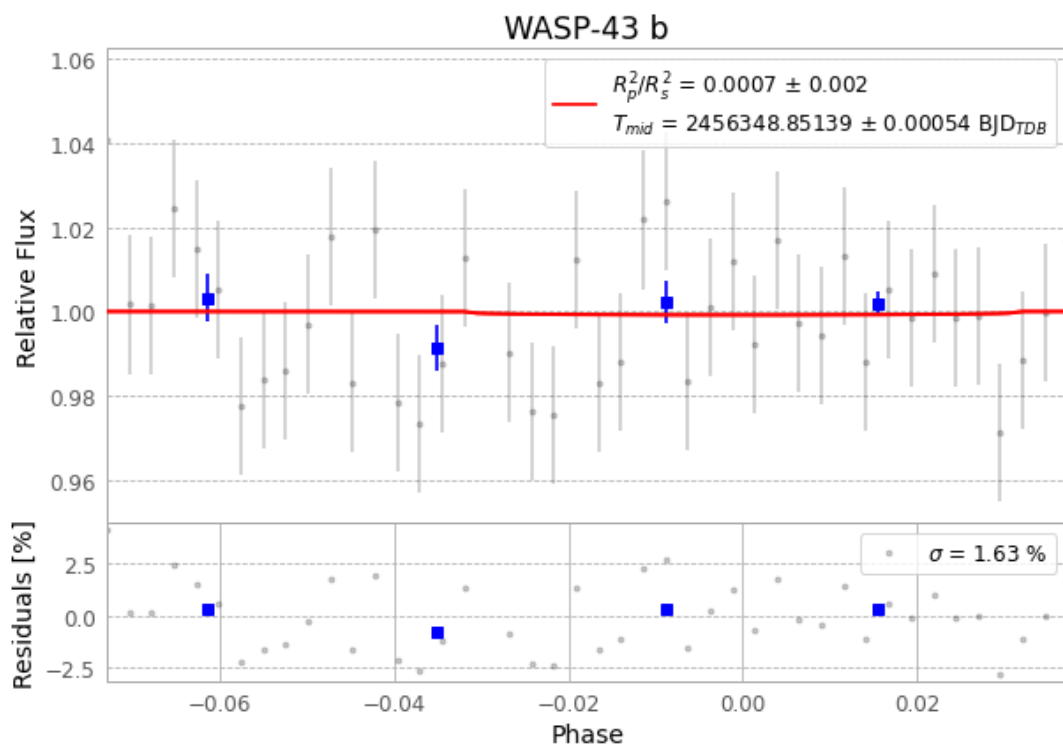


Figure 1 — FinalLightCurve\_WASP-43 b\_2013-02-24.png (SHA-256 b320af5dd01257ff...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [314, 328], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 faec496239d35a7b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit a4253ce

## Data provenance

44 FITS frames (29 MB), WASP-43130225065112.FITS → WASP-43130225090012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130225.log (a61bd9d424e8...), FinalLightCurve\_WASP-43 b\_2013-02-24.png (b320af5dd012...), FinalLightCurve\_WASP-43 b\_2013-02-24.csv (175e36e7803e...)

## Compute resources

Wall time 149.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 00:14Z.